Chapter 11 — Hydrocarbons

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Classification and nomenclature of hydrocarbons

Q.1 Classify hydrocarbons and explain IUPAC nomenclature of alkanes and cycloalkanes.

Hydrocarbons contain only carbon and hydrogen. Aliphatic hydrocarbons include open-chain and cyclic non-aromatic compounds: alkanes are saturated, alkenes and alkynes are unsaturated, and cycloalkanes are saturated rings. Aromatic hydrocarbons are based on benzene or similar stable ring systems.

Alkanes have the general formula C_nH_{2n+2}. To name a branched alkane, choose the longest continuous carbon chain, number it to give substituents the lowest locants, name substituents alphabetically and use di-, tri- for repeated groups. Cycloalkanes have formula C_nH_{2n} and are named by adding the prefix cyclo-.

Textbook check (pp. 238, 243): Table 11.1 prints heptane as C_7H_{10}; the correct alkane formula is C_7H_{16}. Table 11.5 prints cyclopentane as C_4H_{10}; the correct formula is C_5H_{10}.

Important Conceptual Short Questions — Topic 1

1. Why are alkanes called saturated hydrocarbons?

They contain only carbon-carbon single bonds and the maximum possible number of hydrogen atoms for an open-chain hydrocarbon.

2. How does the longest-chain rule determine the parent name?

The longest continuous carbon chain gives the root name, such as pentane or hexane. Shorter chains attached to it are treated as substituents.

3. Why are substituents given the lowest possible numbers?

Lowest locants give a unique and systematic name and make the position of branching clear.

4. How does a cycloalkane differ from an alkene with the same general formula?

A cycloalkane is saturated and contains a ring of single bonds, while an alkene is unsaturated and contains a C=C double bond. They are functional-group/structural isomers.

Topic 2 — Alkanes: properties and free-radical substitution

Q.2 Explain the unreactive nature of alkanes and the mechanism of free-radical substitution.

Alkanes are relatively unreactive towards polar reagents because C–C and C–H bonds are strong and almost non-polar. They do not readily attract electrophiles or nucleophiles. However, they undergo combustion and free-radical substitution with halogens in UV light.

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Chlorination of methane occurs by a chain mechanism. Initiation: $Cl_2 \rightarrow 2Cl\bullet$ under UV. Propagation: $Cl\bullet + CH_4 \rightarrow HCl + CH_3\bullet$; $CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$. Termination occurs when radicals combine, such as $Cl\bullet + Cl\bullet \rightarrow Cl_2$ or $CH_3\bullet + Cl\bullet \rightarrow CH_3Cl$.

Important Conceptual Short Questions — Topic 2

1. Why are alkanes inert towards polar reagents?

Their bonds are strong and nearly non-polar, so they do not provide centres of high positive or negative charge for polar reagents to attack.

2. How does UV light initiate chlorination of methane?

UV light supplies energy for homolytic fission of the Cl–Cl bond, producing chlorine free radicals that start the chain reaction.

3. Why is free-radical substitution a chain reaction?

A radical consumed in one propagation step is regenerated in another, so the reaction can continue through many cycles.

4. How do termination steps stop the reaction?

Two radicals combine to form a stable molecule, removing radicals from the system and ending the chain.

Topic 3 — Alkenes: structure, preparation and electrophilic addition

Q.3 Explain the structure and reactivity of alkenes and describe electrophilic addition reactions.

Alkenes contain a C=C double bond made of one sigma bond and one pi bond. The pi bond is weaker and more exposed, so alkenes are more reactive than alkanes and undergo electrophilic addition. Ethene is planar with bond angles close to 120°.

Alkenes can be prepared by dehydration of alcohols or dehydrohalogenation of alkyl halides. Important addition reactions include hydrogenation to alkanes, hydrohalogenation to alkyl halides, hydration to alcohols, halogenation to dihalides and ozonolysis to carbonyl compounds. Markovnikov’s rule states that in addition of HX to an unsymmetrical alkene, H attaches to the carbon already bearing more hydrogens.

Important Conceptual Short Questions — Topic 3

1. Why are alkenes more reactive than alkanes?

The pi bond in alkenes is weaker and its electron density is more exposed, so electrophiles can attack it readily.

2. How does Markovnikov’s rule predict the major product?

The proton adds to the double-bond carbon with more hydrogens, forming the more stable carbocation. The halide then attaches to the more substituted carbon.

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3. Why does hydration of ethene require an acid catalyst?

The acid provides H^+ to start electrophilic addition and is regenerated at the end, allowing water to add across the double bond.

4. How does ozonolysis help locate a double bond?

Ozone cleaves the C=C bond and forms carbonyl compounds. Identifying these products reveals the position of the original double bond.

Topic 4 — Carbocation stability, inductive effect and isomerism

Q.4 Explain carbocation stability, inductive effect and types of isomerism in hydrocarbons.

Carbocations are electron-deficient carbon species. Their stability increases with the number of alkyl groups: tertiary > secondary > primary > methyl. Alkyl groups show a +I inductive effect, pushing electron density toward the positive carbon and stabilizing it.

Isomers have the same molecular formula but different structures or spatial arrangements. Structural isomerism includes chain, position, functional-group, metamerism and tautomerism. Stereoisomerism includes geometrical cis-trans isomerism and optical isomerism.

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Important Conceptual Short Questions — Topic 4

1. Why is a tertiary carbocation more stable than a primary carbocation?

Three alkyl groups donate electron density by the +I effect and hyperconjugation, spreading the positive charge more effectively than one alkyl group.

2. How does the inductive effect explain Markovnikov addition?

Addition that forms the more substituted carbocation is favoured because alkyl groups stabilize the positive charge. This leads to the Markovnikov product.

3. Why do conjugated dienes show extra stability?

Adjacent p orbitals overlap over several atoms, allowing pi electrons to delocalize. Delocalization lowers the energy of the molecule.

4. How can cis and trans isomers differ in physical properties?

Their different shapes give different dipole moments and packing in the solid state, so boiling points, melting points and solubilities may differ.

Topic 5 — Conjugated dienes and organic redox reactions

Q.5 Explain conjugation in dienes and identify organic oxidation and reduction reactions.

Conjugated dienes have alternate double and single bonds, such as CH_2=CH–CH=CH_2. The adjacent p orbitals overlap over several atoms, allowing pi electrons to delocalize. This delocalization gives extra stability and affects reactivity.

Organic oxidation usually means addition of oxygen or removal of hydrogen. Combustion, ozonolysis and epoxidation are oxidation processes. Organic reduction usually means addition of hydrogen or removal of oxygen, such as hydrogenation of alkenes or reduction of carbonyl compounds to alcohols.

Important Conceptual Short Questions — Topic 5

1. Why are conjugated dienes more stable than isolated dienes?

Delocalization of pi electrons over adjacent p orbitals lowers the energy of the molecule.

2. How can hydrogenation be recognized as reduction?

Hydrogen is added across a multiple bond, increasing the hydrogen content of the organic molecule.

3. Why is combustion an oxidation reaction?

The organic compound reacts with oxygen, forming CO_2 and H_2O and increasing the oxidation state of carbon.

4. How does ozonolysis help in structure determination?

It cleaves a double bond oxidatively and gives carbonyl fragments that reveal the position of the original double bond.

Topic 6 — Electron-withdrawing groups and their effects

Q.6 Explain the inductive and resonance effects of electron-withdrawing groups.

Electron-withdrawing groups pull electron density away from a carbon chain or ring. They may act by the –I inductive effect, as in –NO_2, –CN and halogens, or by resonance, as in –NO_2, –CHO and –COOH. These effects make nearby carbon atoms more electron deficient.

Electron-withdrawing groups stabilize carbanions and destabilize carbocations. They increase the acidity of carboxylic acids by stabilizing the carboxylate ion. On a benzene ring, they deactivate the ring towards electrophilic substitution and often direct incoming groups to the meta position.

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Important Conceptual Short Questions — Topic 6

1. Why do electron-withdrawing groups increase the acidity of carboxylic acids?

They stabilize the negatively charged carboxylate ion, making loss of H^+ easier.

2. How does a –I group affect a carbocation?

It pulls electron density away from the positively charged carbon, increasing electron deficiency and destabilizing the carbocation.

3. Why does a nitro group deactivate benzene towards electrophilic substitution?

It withdraws electron density from the ring by –I and –R effects, making the ring less attractive to electrophiles.

4. How do electron-withdrawing groups stabilize a carbanion?

They disperse or pull away the negative charge, lowering the energy of the carbanion.

Topic 7 — Aromatic hydrocarbons and benzene

Q.7 Explain aromaticity and the characteristic reactions of benzene.

Benzene, C_6H_6, is a planar cyclic molecule with delocalized pi electrons. All six C–C bonds are equal in length, intermediate between single and double bonds. This delocalization gives benzene unusual stability.

Because addition would destroy the stable delocalized system, benzene prefers electrophilic substitution, such as nitration, sulphonation and halogenation in the presence of a catalyst. Aromatic compounds are important starting materials for dyes, drugs, plastics and explosives.

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Important Conceptual Short Questions — Topic 7

1. Why does benzene undergo substitution rather than addition?

Substitution preserves the stable delocalized pi system, while addition would destroy aromaticity and require more energy.

2. How does delocalization explain the equal bond lengths in benzene?

The pi electrons are spread over all six carbon atoms, so each C–C bond has the same partial double-bond character.

3. Why is benzene more stable than a hypothetical cyclohexatriene?

Delocalization of pi electrons lowers the energy of benzene below that expected for three isolated double bonds.

4. Why is benzene resistant to bromine water without a catalyst?

Bromine water would require addition to the ring, which would destroy the stable delocalized pi system. Without a catalyst, benzene does not react readily.